

Sentinels VR Hostage-Rescue Trainer · Module 4

Research Figures & Charts

Sound-Driven Hostage Emotional State — Supporting Visuals for the Final Report

Compiled July 2026

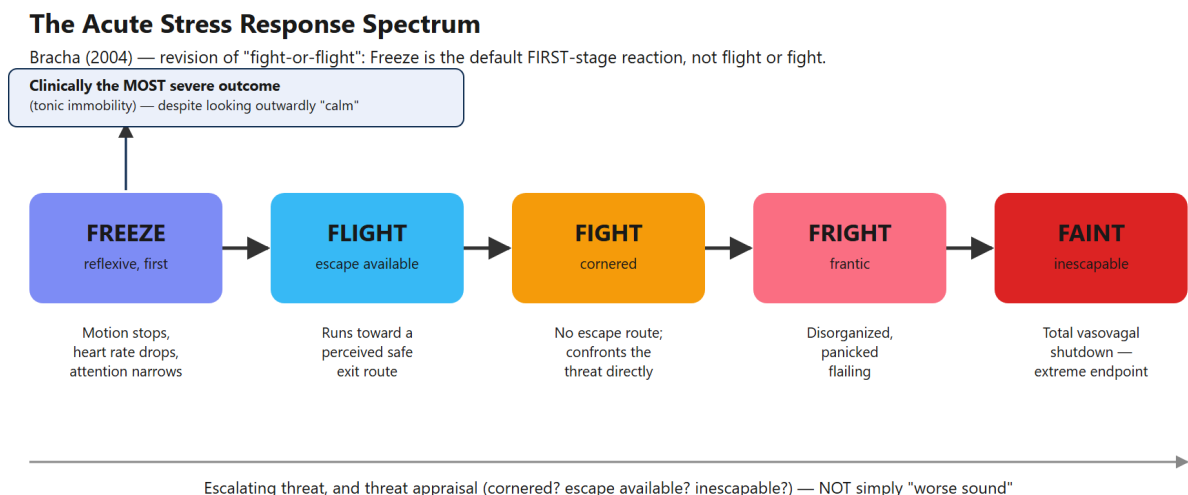
About These Figures

This document collects the charts and diagrams referenced in the hostage sound/emotion research review, for direct use in the Module 4 final report. Every figure here is an original chart built from the underlying research data and findings — none are copied images from the source papers themselves (several of those, e.g. Current Biology, JASA, CNS Spectrums, are commercially copyrighted journal figures not licensed for reproduction). Recreating the data and findings as original charts keeps this document safe to include in your submission while still carrying the same evidentiary content.

Each figure includes its source citation directly in the caption. Full references are listed at the end of this document.

Figure 1 — The Acute Stress Response Spectrum

Bracha (2004) revises the traditional "fight-or-flight" model into a five-stage spectrum, ordered by how a threat is appraised (escape available? cornered? inescapable?) rather than by sound loudness alone. The key, counter-intuitive finding used throughout this project: Freeze is the reflexive FIRST-stage reaction to a sudden threat — and, despite looking calmer than Panic, is clinically the more severe outcome (see Figure 3).



Source: Bracha, H.S. (2004). "Freeze, Flight, Fight, Fright, Faint: Adaptationist Perspectives on the Acute Stress Response Spectrum." *CNS Spectrums*.

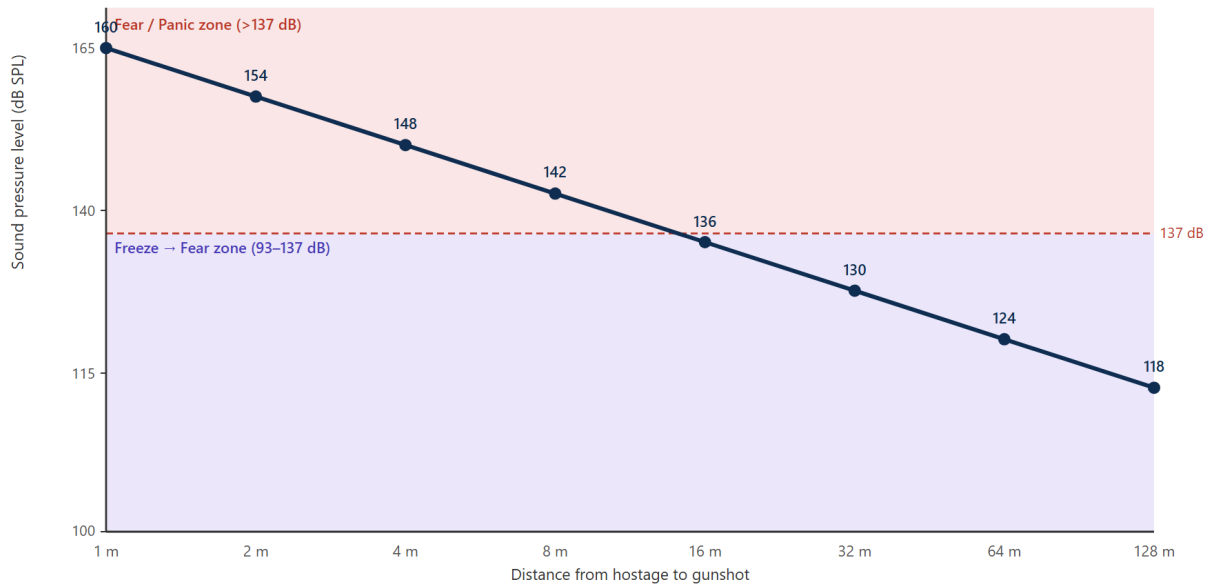
Figure 1. The Freeze → Flight → Fight → Fright → Faint sequence. — Source: Bracha, H.S. (2004), *CNS Spectrums*.

Figure 2 — Gunfire Sound Level vs. Distance

Gunfire SPL is fixed at the source (a rifle produces a consistent -155-165 dB at 1 m every time it fires) and falls off with distance by the standard inverse-square law: -6 dB every time distance doubles. This chart plots that falloff for a representative 160 dB/1 m rifle, shaded by the emotional-state zone each distance band falls into.

Gunfire SPL vs. Distance to Hostage

Typical rifle, 160 dB at 1 m. $SPL(d) = 160 - 20 \cdot \log_{10}(d)$ — the standard inverse-square law



Confirmed against a real measured value: an M-72 rocket at 179 dB at the firer's position dropped to 161 dB at 8 m (18 dB loss) — U.S. Army data, consistent with this line.
Sources: G.R. Price, U.S. Army Research Laboratory, AHAH report · Engineering Toolbox, Inverse-Square Law of Sound.

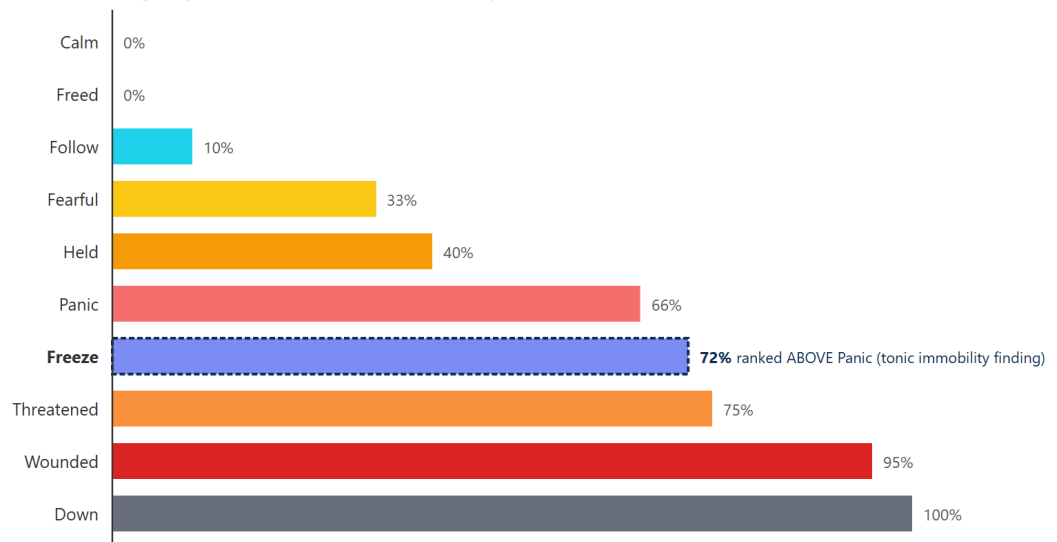
Figure 2. $SPL(d) = 160 - 20 \cdot \log_{10}(d)$, confirmed against a real U.S. Army measurement (M-72 LAW: 179 dB → 161 dB over 8 m). — Sources: G.R. Price, U.S. Army Research Laboratory (AHAH report); Engineering Toolbox.

Figure 3 — Hostage Distress Index by State (Module 4 Model)

This is the actual distress-percentage model used in the Module 4 AAR dashboard (HostageTab.jsx), after the research-driven correction: Freeze (tonic immobility) is scored ABOVE Panic, because peritraumatic tonic immobility is clinically shown to predict worse PTSD outcomes than active panic — the opposite of what its outwardly "calm" appearance suggests.

Hostage Distress Index by State

Module 4 AAR dashboard model (HostageTab.jsx STATE_INFO), after the tonic-immobility research correction



Freeze raised from 50% to 72% (above Panic's 66%); peritraumatic tonic immobility predicts WORSE PTSD outcomes than active panic, despite outwardly looking calmer. Sources: ScienceDirect 2019 (tonic immobility & PTSD recovery); Brazilian police officers TI study; Arnal et al. 2015, Current Biology.

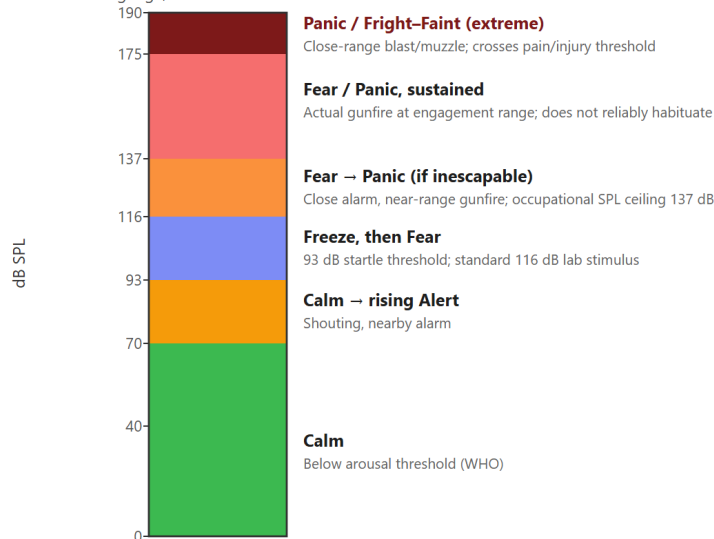
Figure 3. Distress % per hostage state, as implemented in STATE_INFO. — Sources: ScienceDirect (2019) tonic immobility & PTSD recovery; Brazilian police officers TI study; Arnal et al. (2015), Current Biology.

Figure 4 — General Sound Level → Emotional State Reference

A broader reference gauge (not limited to gunfire) mapping SPL bands to the expected emotional-state response of an average person, spanning ambient quiet up to close-range blast overpressure.

Sound Pressure Level → Emotional State (Any Sound)

General reference gauge, 0–190 dB SPL



Sources: WHO Environmental Noise Guidelines · Acoustic Startle Chamber methodology · Fear-potentiated startle review (PMC6162305)
Firearms occupational SPL limit (JASA) · Blast-exposed failure-to-habituate study (PMC6387566) · Firearms/blast injury threshold (PMC9450760)

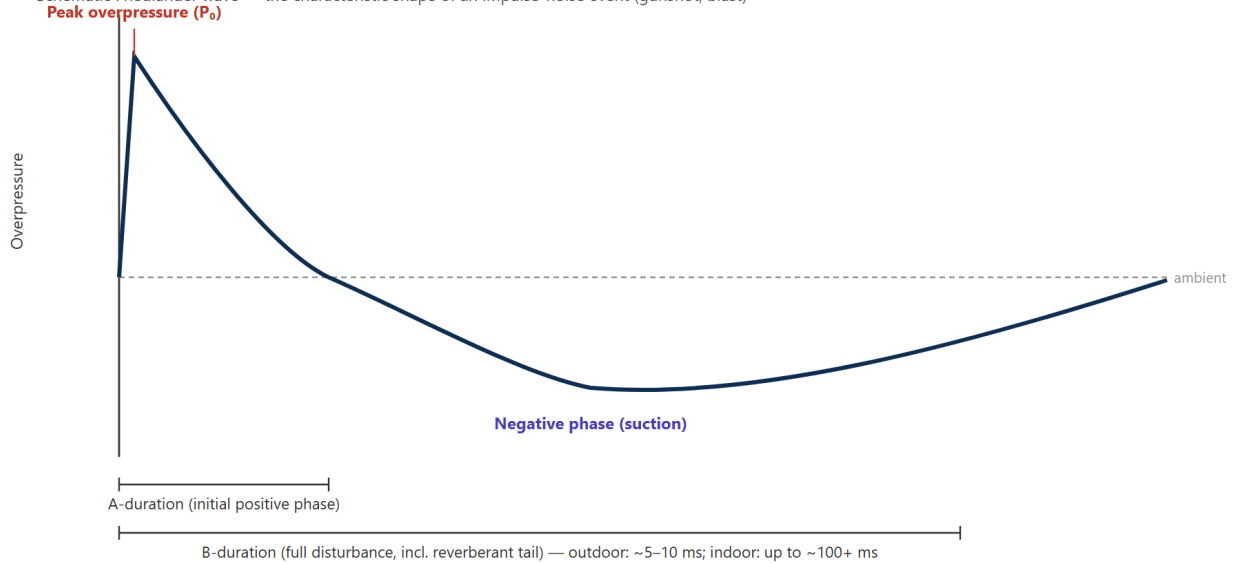
Figure 4. 0–190 dB SPL reference gauge with state bands. — Sources: WHO Environmental Noise Guidelines; acoustic startle chamber methodology; PMC6162305; JASA; PMC6387566; PMC9450760.

Figure 5 — Idealized Gunfire / Blast Pressure Waveform

Loudness (SPL) is only part of the picture — the SHAPE of the pressure wave over time also matters. Gunfire and blast events produce a distinctive "Friedlander wave": an almost-instantaneous rise to peak pressure, followed by an exponential decay through ambient into a longer, shallower negative (suction) phase. This rapid rise time is what makes gunfire trigger a reflexive startle rather than a slower cardiac orienting response.

Idealized Gunfire / Blast Pressure Waveform

Schematic Friedlander wave — the characteristic shape of an impulse-noise event (gunshot, blast)



Schematic, after Price, G.R., U.S. Army Research Laboratory, "Weapon Noise Exposure of the Human Ear Analyzed with the AHAAH Model" (public-domain U.S. Government work).

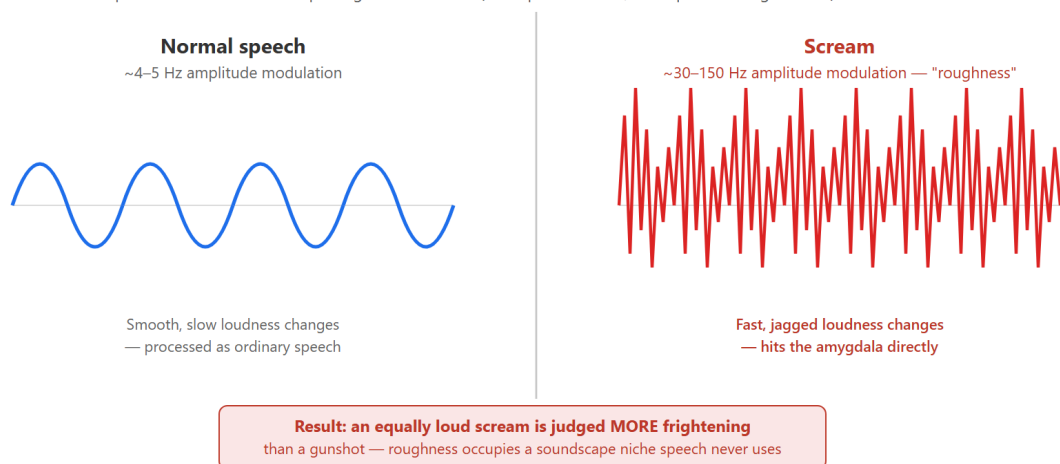
Figure 5. Schematic Friedlander waveform (illustrative, not measured data). — Source: Price, G.R., U.S. Army Research Laboratory — public-domain U.S. Government work.

Figure 6 — Why a Scream Outranks a Gunshot

Arnal et al. (2015) found that screams carry a distinctive acoustic property — "roughness," extremely fast amplitude modulation (30–150 Hz) — that normal speech never uses (speech modulates at ~4–5 Hz). This roughness activates the amygdala's fear circuitry directly, more strongly than an equally loud sound without it. This is the basis for weighting a captor's vocal threat ("Threatened", 75%) above plain gunfire-triggered Panic (66%) in the distress model.

Why a Scream Outranks a Gunshot

"Roughness" — fast amplitude modulation — is a privileged fear channel (concept illustration, not reproduced figure data)



Source: Arnal, L. et al. (2015). "Human Screams Occupy a Privileged Niche in the Communication Soundscape." *Current Biology*.

Figure 6. Slow speech-envelope modulation vs. fast scream-envelope "roughness" (concept illustration). — Source: Arnal, L. et al. (2015), *Current Biology*.

References

Bracha, H.S. (2004). "Freeze, Flight, Fight, Fright, Faint: Adaptationist Perspectives on the Acute Stress Response Spectrum." *CNS Spectrums*.

Price, G.R. U.S. Army Research Laboratory. "Weapon Noise Exposure of the Human Ear Analyzed with the AHAAH Model."

Arnal, L. et al. (2015). "Human Screams Occupy a Privileged Niche in the Communication Soundscape." *Current Biology*.

Tonic immobility predicts poorer PTSD recovery. *ScienceDirect* (2019).

Peritraumatic tonic immobility and PTSD symptom severity in Brazilian police officers: a prospective study.

Fear-potentiated startle review. PMC6162305.

Blast-exposed failure-to-habituate startle-gating study. PMC6387566.

Acoustic and psychoacoustic analysis of police firearms noise. PMC9450760.

Noise of military weapons, ground vehicles, planes and ships. *Journal of the Acoustical Society of America (JASA)*.

WHO Environmental Noise Guidelines for the European Region.

Engineering Toolbox. "Sound Propagation — the Inverse Square Law."

Companion document: HOSTAGE_EMOTION_SOUND_RESEARCH.pdf (full research review with dB-by-distance tables). Full URLs for every source are listed there.